

THE MACROECONOMIC EFFECTS OF LIFTING BARRIERS TO MIGRATION

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Stockman

STARK URBAN-RURAL GAP IN LOW-INCOME COUNTRIES



Pictures show a rural Kenyan home and Nairobi, the capital city of Kenya. Source: Jocelyn Diaz (left) and Catalin Marin (right).

RETURNS TO MIGRATION ARE HIGH

- huge urban–rural gaps in income
 - Kenya: **3.3x urban–rural productivity gap** (URPG)
 - large **marginal income returns** to migration (Bryan, Chowdhury, Mobarak, 2014)
- urbanization remains limited
- growing randomized controlled trial evidence on why!

WHY DO GAPS PERSIST AND WHAT CAN WE DO? RCT EVIDENCE

- migration costs, fin. constraints (*Bryan, Chowdhury, Mobarak, 2014*)
 - provide cash to cover migration cost
- unemployment risk (*Miner, 2026*)
 - provide urban unemployment insurance
- underestimation of urban income (*Baseler, 2023; Barnett-Howell et al., 2026*)
 - provide information about urban earnings

which barriers are costliest?
which interventions are most effective?

- unclear because experiments have
 - different environment and sampling
 - different intervention design and measurement
 - models tailored to specific experiment

THIS PAPER: A UNIFIED MODEL OF MIGRATION BARRIERS AND INTERVENTIONS

- build and estimate a **quantitative general equilibrium model** of internal migration to match moments from Kenyan data
- reproduce **several prominent experiments** in Kenya within a single environment
 - approximate sampling, intervention, and measurement of each without targeting treatment effects
 - **unconditional cash transfers** (*Egger et al., 2022; Miner, 2026*) ► *Egger* ► *Miner*
 - Egger: consumption ↑, no migration effect; Miner: migration ↑
 - **unemployment insurance** (*Miner, 2026*) ► *details*
 - migration ↑
 - **information provision** (*Barnett-Howell et al., 2026*) ► *design* ► *findings*
 - migration and income ↑; group delivery: no income effect

PREVIEW: RESULTS

- model matches some migration **treatment effects**, misses others
- **long-run effects** can differ substantially from experimental endlines
 - large cash transfer: no migration effect in SR, becomes significant in LR
 - UI and information: effects continue to grow after endline
- **national rollout** of UI, mentor intervention modestly effective
 - urbanization: 30.5% → **35.3–35.6%**
 - URP: 3.31 → **3.18–3.20**
- decompose the **urban–rural income gap** into contributions from different frictions:
 - information is the strongest explicit friction: explains **19%** of the URP
 - residual migration costs explain **87%** of the URP

MODEL

MODEL OVERVIEW

Aiyagari-style dynamic heterogeneous agent model with migration

REGIONS

- Rural (R) and Urban (U)

HETEROGENEOUS HOUSEHOLDS CHOOSE

- consumption c
- savings a'
- location l'

HOUSEHOLDS HETEROGENEOUS IN

- rural, urban productivity z_R, z_U
- location l
- urban experience x
- employment status e
- assets a

FRICTIONS

- borrowing constraint: $a \geq \underline{a}$
- information friction:
 - rural households perceive $\tilde{w}_U = \gamma w_U$
- urban unemployment risk:
 - job separation λ_s
 - job finding λ_f
- migration costs: pay m_c when moving
- migration disutility:
 - the inexperienced discount urban consumption by m_u

PREFERENCES AND EXPERIENCE

- CRRA utility over effective consumption:

$$u(c, l, x) = \frac{\tilde{c}^{1-\sigma} - 1}{1 - \sigma}, \quad \tilde{c} = \frac{c}{1 + m_u \cdot \mathbf{1}(l = U, x = 0)}$$

- m_u : consumption-equivalent disutility for inexperienced urban migrants
 - inexperienced migrants in urban region discount consumption
- urban experience $x \in \{0, 1\}$:
 - gain experience ($x' = 1$) with prob. π_x per period in urban, stay experienced
 - lose experience ($x' = 0$) with prob. π_{-x} per period in rural, stay inexperienced
 - as in *Lagakos, Mobarak, Waugh (2023)*

INCOME, PRODUCTIVITY, AND EMPLOYMENT

budget constraint: $c + a' + \mathbf{1}(l = R, l' = U) \cdot m_c = (1 + r)a + y(e, l, z_R, z_U)$

EMPLOYMENT

- urban: stochastic employment e
 - $e = 1 \rightarrow 0$: job separation rate λ_s
 - $e = 0 \rightarrow 1$: job finding rate λ_f
- rural: always employed ($e = 1$)

INCOME

$$y = \begin{cases} w_R z_R & \text{rural} \\ w_U z_U & \text{urban employed} \\ b & \text{urban unemployed} \end{cases}$$

rural beliefs: $\tilde{w}_U = \gamma w_U$, $\gamma \in (0, 1]$ ► *details*

MIGRATION COST

- pay m_c when moving $R \rightarrow U$

BIVARIATE AR(1) PRODUCTIVITY PROCESS

$$\log z'_R = \rho_R \log z_R + \epsilon'_R$$

$$\log z'_U = \rho_U \log z_U + \epsilon'_U$$

$$(\epsilon'_R, \epsilon'_U) \sim N(\mathbf{0}, \Sigma)$$

- ρ_ϵ : cross-correlation of shocks

ASSETS

- save in one-period risk-free bond a'
 - return r
- borrowing constraint: $a' \geq \underline{a}$

► *production and GE*

ESTIMATION

ESTIMATION

ASSIGNED PARAMETERS

t	period length
β	discount factor
σ	risk aversion
α	capital share
δ	depreciation
A_R	rural TFP
λ_s	job separation
λ_f	job finding
m_u	migration disutility
π_x	experience gain
π_{-x}	experience loss
γ	info friction
b	UI benefit
$\underline{a}$	borrowing limit

ESTIMATED PARAMETERS

A_U	urban TFP
ρ_R	rural persistence
ρ_U	urban persistence
$\sigma_{\epsilon,R}$	rural shock SD
$\sigma_{\epsilon,U}$	urban shock SD
ρ_{ϵ}	shock correlation
m_c	migration cost

$t = 6$ months

length of one model period

- maximizes available data moments for estimation

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$\beta = 0.98$
discount factor

$$V_{l \rightarrow l'} = \max_{a' \geq \underline{a}} u(c, R, x) \\ + \beta \mathbb{E}[V_{l'}(a', z'_R, z'_U, x', e', \gamma)]$$

- standard for 6-month period length (from 0.96 for annual)

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$$\sigma = 2$$

coefficient of risk aversion in CRRA utility

$$u(c, l, x) = \frac{\tilde{c}^{1-\sigma} - 1}{1 - \sigma}$$

- standard

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$$\alpha = 0.33$$

capital share in production

$$Y_l = A_l K_l^{\alpha} L_l^{1-\alpha}$$

- labor share: 0.67
- standard

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$$\delta = 0.051$$

capital depreciation rate

$$r + \delta = \alpha A_l (K_l / L_l)^{\alpha-1}$$

- standard for 6-month period length (from 0.10 for annual)

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$$A_R = 1$$

rural total factor productivity

$$Y_R = A_R K_R^{\alpha} L_R^{1-\alpha}$$

- normalized

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$\lambda_s = 0.06$
urban job separation rate

$$\Pr(e' = 0 \mid e = 1, l = U) = \lambda_s, \quad \Pr(e' = 1 \mid e = 0, l = U) = \lambda_f$$

- **target moment:** 6-month unemployment hazard rate for urban migrants
- **data:** survey in *Barnett-Howell et al. (2026)*

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m_c	migration cost

$\lambda_f = 0.42$
urban job finding rate

$$\Pr(e' = 0 \mid e = 1, l = U) = \lambda_s, \quad \Pr(e' = 1 \mid e = 0, l = U) = \lambda_f$$

- **target moment:** 6-month employment hazard rate for urban migrants
- **data:** survey in *Barnett-Howell et al. (2026)*

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m_c	migration cost

$$m_u = 0.34$$

consumption-equivalent utility cost of migration

$$\tilde{c} = \frac{c}{1 + m_u \cdot \mathbf{1}(l = U, x = 0)}$$

- **target moment:** share of income migrants are willing to give up to move back to rural, among those who are willing to give up some income to move back (“rural preference”)
 - survey asked migrants what rural income would make them willing to move back
- **data:** survey in *Barnett-Howell et al. (2026)*

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$\pi_x = 0.45$
urban experience gaining rate

$$\Pr(x' = 1 \mid l = U, x = 0) = \pi_x, \quad \Pr(x' = 0 \mid l = R, x = 1) = \pi_{-x}$$

- **target moment:** among rural households with rural preference, what share switches to urban preference within 6 months after migrating
 - location preference: require higher income to move to the other location
- **data:** survey in *Barnett-Howell et al. (2026)*

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$$\pi_{-x} = 0.68$$

urban experience losing rate

$$\Pr(x' = 1 \mid l = U, x = 0) = \pi_x, \quad \Pr(x' = 0 \mid l = R, x = 1) = \pi_{-x}$$

- **target moment:** among urban households with urban preference, what share switches to rural preference within 6 months after migrating
 - location preference: require higher income to move to the other location
- **data:** survey in *Barnett-Howell et al. (2026)*

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m_c	migration cost

$$\gamma = 0.56$$

understatement of urban income by rural households

$$\tilde{w}_U = \gamma w_U$$

- **target moment:** avg expected Nairobi income by rural households as a share of actual Nairobi income, conditional on demographics
- **data:** *Barnett-Howell et al. (2026)*

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$\sigma_{\epsilon,U}$	urban shock SD
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$$b = 0$$

urban unemployment benefit

$$y(e, U, z_R, z_U) = \begin{cases} w_U z_U & e = 1 \\ b & e = 0 \end{cases}$$

- no formal unemployment insurance in Kenya at baseline

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$\sigma_{\epsilon,U}$	urban shock SD
ρ_{ϵ}	shock correlation
m_c	migration cost

$$\underline{a} = 0$$

borrowing limit

$$c + a' + \mathbf{1}(l = R, l' = U) \cdot m_c = (1 + r)a + y(e, l, z_R, z_U), \quad a' \geq \underline{a}$$

- no borrowing allowed at baseline

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$\sigma_{\epsilon,U}$	urban shock SD
ρ_{ϵ}	shock correlation
m_c	migration cost

$$A_U = 2.02$$

urban total factor productivity

$$Y_U = A_U K_U^{\alpha} L_U^{1-\alpha}$$

- **target moment:** exponentiated difference in mean log urban and rural incomes
- **data:** Kenya Integrated Household Budget Survey (KIHBS)

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m_c	migration cost

$$\rho_R = 0.68$$

persistence of rural productivity

$$\log Z'_R = \rho_R \log Z_R + \epsilon'_R$$

$$\log Z'_U = \rho_U \log Z_U + \epsilon'_U$$

$$SD(\epsilon'_R) = \sigma_{\epsilon,R}, \quad SD(\epsilon'_U) = \sigma_{\epsilon,U}$$

$$\text{Corr}(\epsilon'_R, \epsilon'_U) = \rho_\epsilon$$

- **target moment:** standard deviation of log rural income
- **data:** Kenya Integrated Household Budget Survey (KIHBS)

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ρ_U	urban persistence
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$\sigma_{\epsilon,U}$	urban shock SD
ρ_ϵ	shock correlation
m_c	migration cost

$$\rho_U = 0.70$$

persistence of urban productivity

$$\log Z'_R = \rho_R \log Z_R + \epsilon'_R$$

$$\log Z'_U = \rho_U \log Z_U + \epsilon'_U$$

$$SD(\epsilon'_R) = \sigma_{\epsilon,R}, \quad SD(\epsilon'_U) = \sigma_{\epsilon,U}$$

$$\text{Corr}(\epsilon'_R, \epsilon'_U) = \rho_\epsilon$$

- **target moment:** standard deviation of log urban income
- **data:** Kenya Integrated Household Budget Survey (KIHBS)

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ρ_ϵ	shock correlation
m_c	migration cost

$$\sigma_{\epsilon,R} = 1.12$$

volatility of rural productivity shocks

$$\log Z'_R = \rho_R \log Z_R + \epsilon'_R$$

$$\log Z'_U = \rho_U \log Z_U + \epsilon'_U$$

$$SD(\epsilon'_R) = \sigma_{\epsilon,R}, \quad SD(\epsilon'_U) = \sigma_{\epsilon,U}$$

$$\text{Corr}(\epsilon'_R, \epsilon'_U) = \rho_\epsilon$$

- **target moment:** standard deviation of residual rural log income after controlling for persistence
- **data:** survey in *Barnett-Howell et al. (2026)*

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ρ_ϵ	shock correlation
m_c	migration cost

$$\sigma_{\epsilon,U} = 1.04$$

volatility of urban productivity shocks

$$\log Z'_R = \rho_R \log Z_R + \epsilon'_R$$

$$\log Z'_U = \rho_U \log Z_U + \epsilon'_U$$

$$SD(\epsilon'_R) = \sigma_{\epsilon,R}, \quad SD(\epsilon'_U) = \sigma_{\epsilon,U}$$

$$\text{Corr}(\epsilon'_R, \epsilon'_U) = \rho_\epsilon$$

- **target moment:** standard deviation of residual urban log income after controlling for persistence
- **data:** survey in *Barnett-Howell et al. (2026)*

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ρ_ϵ	shock correlation
m_c	migration cost

$$\rho_\epsilon = 0.01$$

cross-location correlation of productivity shocks within a household.

$$\log Z'_R = \rho_R \log Z_R + \epsilon'_R$$

$$\log Z'_U = \rho_U \log Z_U + \epsilon'_U$$

$$SD(\epsilon'_R) = \sigma_{\epsilon,R}, \quad SD(\epsilon'_U) = \sigma_{\epsilon,U}$$

$$\text{Corr}(\epsilon'_R, \epsilon'_U) = \rho_\epsilon$$

- **target moment:** slope from regressing urban log income on rural log income for similar households (clustered by observables)
- **model moment:** slope from regressing potential urban log income on potential rural log income within household
- **data:** *Barnett-Howell et al. (2026)*

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ρ_ϵ	shock correlation
m_c	migration cost

$$m_c = 65$$

monetary migration cost

$$c + a' + 1(l = R, l' = U) m_c = (1 + r)a + y(e, l, z_R, z_U)$$

- **target moment:** share of Kenyan population living in cities
- **model moment:** share of households in urban region
- **data:** Kenya National Bureau of Statistics

ESTIMATED PARAMETERS AND TARGET MOMENTS

Parameter	Value	Target moment	Moment value	
			Data	Model
A_U	2.02	URPG	3.28	3.31
ρ_R	0.68	Standard deviation of log rural income	1.57	1.58
ρ_U	0.70	Standard deviation of log urban income	1.46	1.47
$\sigma_{\epsilon,R}$	1.12	Standard deviation of residual rural log income	1.14	1.15
$\sigma_{\epsilon,U}$	1.04	Standard deviation of residual urban log income	1.05	1.06
ρ_{ϵ}	0.01	Slope of urban vs rural log incomes	0.06	0.07
m_C	65	Urban population share	0.32	0.30

REPRODUCING EXPERIMENTS

REPRODUCING EXPERIMENTS: STRATEGY

- for each RCT intervention, create a model representation of:
 - sampling
 - treatment
 - matching some metric of treatment intensity, *not* treatment effects
 - leaves treatment effects to serve as non-targeted moments
 - measurement

CASH TRANSFER: EGGER ET AL. (2022)

SAMPLING	TREATMENT	MEASUREMENT
RCT • 1 rural Kenyan county • half of villages treated • eligibility: thatched roof (1/3 of households)	RCT • one-time unconditional transfer • 75% of mean annual expenditure	RCT • endline: 19 months after transfer • migration and household expenditure, treatment vs control
MODEL • eligibility: poorest 1/3 of rural households by assets • randomly treat half	MODEL • one-time asset transfer • 75% of mean rural annual consumption	MODEL • endline: 3 periods (18 months) • migration: urban residence at endline • consumption

CASH TRANSFER: EGGER ET AL. (2022) RESULTS

Treatment arm	Migration Rate Change		Consumption Change	
	RCT	MODEL	RCT	MODEL
Cash transfer	1.0 pp (1.0)	0.0 pp	11.6% ^{***} (2.4)	14.3%

- model matches negligible short-run migration response, magnitude of consumption response

► *alternative experiments*

UNEMPLOYMENT INSURANCE AND CASH TRANSFER: MINER (2026)

SAMPLING	TREATMENT	MEASUREMENT
RCT • male rural workers aged 18–30 in 1 rural county • most had prior migration experience • sampled 11.7 workers per village MODEL • sample 11.7% of rural households • prioritize experienced rural households	RCT • UI: up to 15 days at average urban wage (60 USD) • cash: one-time 50 USD transfer MODEL • UI: b matching 15 days of avg urban wage • cash: asset transfer, 50/60 of UI amount	RCT • endline: about 6 months after treatment • Nairobi residence and monthly wages plus self-employment profits MODEL • endline: 1 period (6 months) • urban residence and income changes relative to rural baseline income

UNEMPLOYMENT INSURANCE AND CASH TRANSFER: MINER (2026) RESULTS

Treatment arm	Migration Rate Change		Income Change	
	RCT	MODEL	RCT	MODEL
Unemployment insurance	8.2 pp ^{***} (1.8)	2.1 pp	3.0% (6.5)	3.3%
Cash transfer	4.1 pp ^{**} (1.8)	0.1 pp	-5.5% (7.8)	0.4%

- model correctly predicts larger response to UI than cash
- model understates migration response to UI and cash
- model predicts modest income gains, within the 68% CI of insignificant RCT effect

INFORMATION PROVISION: BARNETT-HOWELL ET AL. (2026)

SAMPLING	TREATMENT	MEASUREMENT
RCT • 17,000 rural households in 5 counties • 30% of households within sampled villages treated MODEL • randomly treat 30% of rural households	RCT • information: beliefs $\uparrow$ 4% • mentor: beliefs $\uparrow$ 10% • group: beliefs $\uparrow$ 12% MODEL • information: $\gamma_{treated} = 1.04\gamma$ • mentor: $\gamma_{treated} = 1.10\gamma$ • group: experienced treated households set to $\gamma = 1$; inexperienced γ chosen to match average belief gain	RCT • endline: about 16 months after treatment • any Nairobi migrant and household income MODEL • endline: 3 periods (18 months) • urban residence and income changes relative to rural baseline income

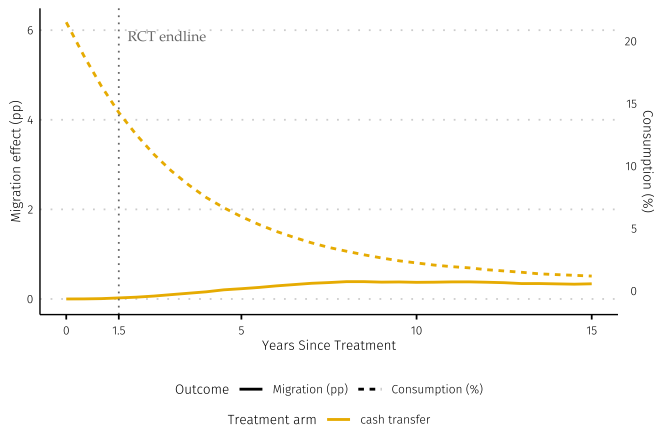
INFORMATION PROVISION: RESULTS

Treatment arm	Migration Rate Change		Income Change	
	RCT	MODEL	RCT	MODEL
Information	1.8 pp [*] (1.0)	0.9 pp	8.9% ^{**} (3.7)	1.2%
Mentor	2.4 pp ^{**} (1.2)	2.5 pp	9.3% ^{**} (4.2)	3.4%
Group	2.2 pp [*] (1.3)	4.3 pp	-2.8% (5.1)	4.9%

- information, mentor: model matches migration, understates income responses
- group: model overpredicts both migration and income
 - group more effective because of greater change in beliefs
 - heterogeneous engagement of experienced/inexperienced households not strong enough to suppress the income effect

QUANTITATIVE RESULTS

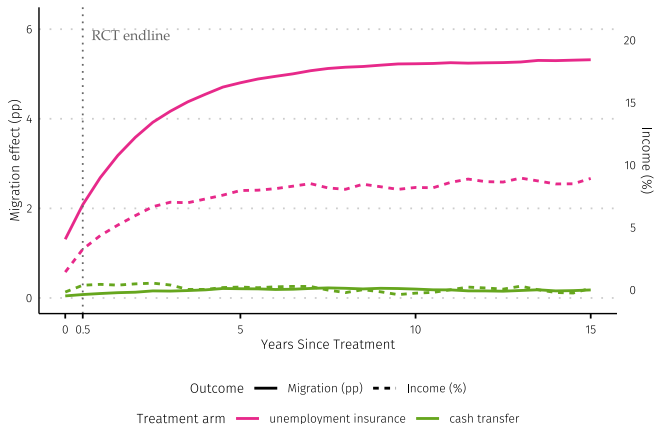
LONG-RUN EFFECTS: CASH TRANSFER, EGGER ET AL. (2022)



- consumption effects decay slowly as the transfer is spent
- migration is negligible at endline, but builds to 0.4 pp after 15 years
 - as more and more rural households experience shocks making migration attractive
- little migration response because targeted low-asset households: most can't afford the move even after the transfer

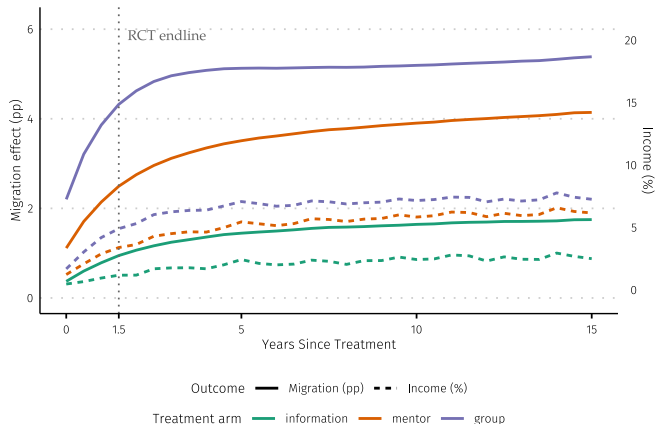
► *alternative experiments*

LONG-RUN EFFECTS: UI AND CASH TRANSFER, MINER (2026)



- UI: about 40% of the 5-year migration effect is realized by endline
- cash: migration builds for several years, then slowly decays

LONG-RUN EFFECTS: INFORMATION PROVISION



- roughly one-half to four-fifths of long-run migration effects are realized by endline
- gains continue growing for a few years as favorable shocks make migration attractive

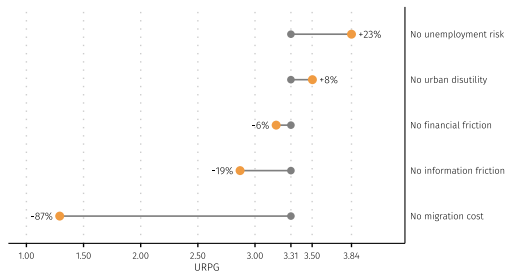
NATIONAL ROLLOUT

- apply info and UI treatments to all households

Scenario	Urban Share	URPG	Consumption
Baseline	30.5%	3.314	1.000
Barnett-Howell et al.: Individual	32.6%	3.242	1.018
Barnett-Howell et al.: Mentor	35.3%	3.178	1.040
Miner: Unemployment insurance	35.6%	3.197	1.064

- all interventions modestly boost urbanization, lower the urban–rural income gap
- mentor and UI generate the largest urbanization gains
 - but cost not equalized! UI much more expensive

URBAN-RURAL INCOME GAP DECOMPOSITION



Scenario	Urban Share	URPG	Consumption
Baseline	30.5%	3.314	1.000
No information friction	46.0%	2.869	1.078
No financial friction	36.2%	3.185	1.122
No unemployment risk	33.5%	3.844	1.212
No urban disutility	27.9%	3.502	0.973
No migration cost	77.3%	1.291	1.640

- information is the most important explicitly modeled friction for the income gap
- residual migration costs are the dominant barrier in the model

CONCLUSION

SUMMARY AND NEXT STEPS

- **model:** model of rural–urban migration
 - info frictions, borrowing constraints, urban risk, migration costs
- **experiments:** reproduce several RCTs in one framework
 - matches Egger et al. cash transfer well
 - matches UI vs cash ranking in Miner, understates migration effect
 - matches migration effect of info, mentor arms in Barnett-Howell et al., understates income
 - overstates migration, income effects of group arm in Barnett-Howell et al.
- **quantitative results:**
 - national UI generates the largest urbanization gain
 - info frictions most important explicit friction behind URP
 - most of the URP is unexplained by explicit frictions, picked up by residual costs
- **further applications of the framework:**
 - compare cost-effectiveness of different interventions
 - slack mechanism for realistic local demand spillovers

CASH TRANSFER: EGGER ET AL. (2022)

Egger, Haushofer, Miguel, Niehaus, Walker (2022)

- **sample:** 10,500 poor households across 653 rural villages
 - randomly sampled from 653 villages within 1 county
- **intervention:** one-time huge unconditional cash transfer
 - 75% of mean annual household expenditure among recipients
 - 16% of annual village GDP
- **findings:**
 - expenditure $\uparrow$ 12%
 - no migration effect
 - large local spillovers: consumption $\uparrow$ similarly among treated and untreated
 - no change in local labor supply
 - local firms: sales $\uparrow$, wages $\uparrow$
- **requirements for model:**
 - financial frictions

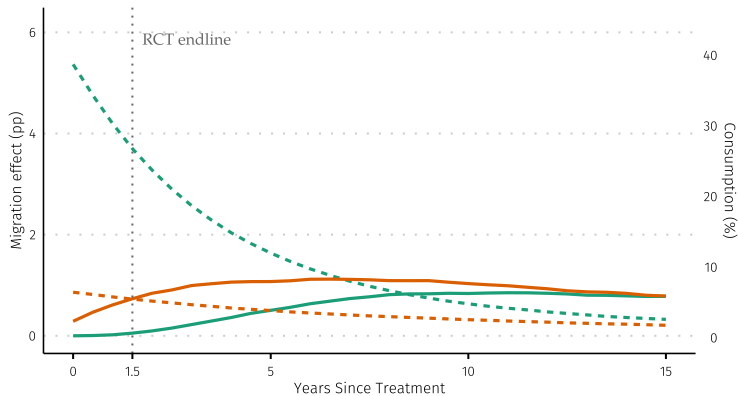
CASH TRANSFER: EGGER ET AL. (2022), ALTERNATIVE EXPERIMENTS

Model treatment effects at endline (18 months)

Sampling	Migration Rate Change			Consumption Change		
	LOW-ASSET, 1x	REPRESENTATIVE, 1x	LOW-ASSET, 2x	LOW-ASSET, 1x	REPRESENTATIVE, 1x	LOW-ASSET, 2x
Cash transfer	0.0 pp	0.7 pp	0.1 pp	14.3%	5.5%	26.8%

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LONG-RUN EFFECTS: EGGER ET AL. (2022), ALTERNATIVE EXPERIMENTS



Outcome — Migration (pp) - - Consumption (%)

Treatment arm — Representative, 1x — Low-asset, 2x

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[endline effects](#)

CASH TRANSFER: MINER (2026)

Miner (2026)

- **sample:** about 1,300 young male workers
 - randomly sampled from villages within 1 county
- **intervention:** one-time unconditional cash transfer
- **findings:**
 - current Nairobi residence $\uparrow$ 4.1 p.p. (SE 1.8)
 - income: -5.5% (SE 7.8%), insignificant

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UNEMPLOYMENT INSURANCE: MINER (2026)

Miner (2026)

- **sample:** about 1,300 young male workers
 - randomly sampled from villages within 1 county
- **intervention:** offer daily UI in Nairobi
 - ~ daily urban wage
 - up to 15 days
- **findings:**
 - current Nairobi residence $\uparrow$ 8.2 p.p. (SE 1.8)
 - income: 3.0% (SE 6.5%), insignificant
- **requirements for model:**
 - unemployment risk

INFORMATION PROVISION: BARNETT-HOWELL ET AL. (2026)

Barnett-Howell, Baseler, Ginn, Gordeev (2026)

- **sample:** 17,000 rural households
 - representatively sampled from 5 rural counties
- **intervention:** information about Nairobi income, labor market
- treatment arms:
 - **information:** individual delivery
 - **mentor:** individual + matched with urban mentor
 - **group:** delivery in village-wide meeting with group discussion

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INFORMATION PROVISION: BARNETT-HOWELL ET AL. (2026)

Barnett-Howell, Baseler, Ginn, Gordeev (2026)

- **findings:**
 - villagers underestimate conditional urban income by 44%
 - treatment → migration rate ↑ 2 p.p.
 - individual/mentor treatment → income ↑ approximately 9%
 - group treatment → no significant income change
 - group treatment primarily engages experienced migrants, who have low marginal returns (because of low migration cost)
 - large economic spillovers within village
- **requirements for model:**
 - information friction
 - heterogeneous migration cost (experience?) affecting marginal returns

INFORMATION FRICTION

- rural households perceive urban wage as $\tilde{w}_U = \gamma w_U$, where $\gamma \in (0, 1]$
 - $\gamma < 1$: rural households underestimate potential urban income
 - perceived income:

$$\tilde{y}(e, l, z_R, z_U, \gamma) = \begin{cases} w_R z_R & \text{if } l = R \\ \gamma w_U z_U & \text{if } l = U \text{ and } e = 1 \\ b & \text{if } l = U \text{ and } e = 0 \end{cases}$$

- as in *Barnett-Howell et al. (2026)*
- rural→urban migrants are surprised by true urban wage
- urban households revert to misperception if they return to rural ► *belief dynamics*
 - otherwise everyone becomes well-informed in stationary equilibrium

PRODUCTION AND GENERAL EQUILIBRIUM

Production

- production in each location l :

$$Y_l = A_l K_l^\alpha L_l^{1-\alpha}$$

- competitive input markets:

$$w_l = (1 - \alpha) A_l (K_l / L_l)^\alpha$$

$$r + \delta = \alpha A_l (K_l / L_l)^{\alpha-1}$$

- perfect capital mobility
 - r equalized across regions

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Equilibrium

1. household optimization given prices
 - rural households solve V_R with $\gamma < 1$
 - urban households solve V_U with $\gamma = 1$
2. firm optimization
3. market clearing:
 - labor: $L_l = \int z_l d\mu$
 - capital: $K_R + K_U = \int a d\mu$
4. stationary distribution

► *rural household problem* ► *urban household problem*

RECURSIVE FORMULATION: RURAL HOUSEHOLD

rural household stays or migrates to urban:

$$V_R(a, z_R, z_U, x, \gamma) = \max \{ V_{R \rightarrow R}, V_{R \rightarrow U} \}$$

STAY IN RURAL

$$V_{R \rightarrow R} = \max_{a' \geq \underline{a}} u(c, R, x) + \beta \mathbb{E}[V_R(a', z'_R, z'_U, x', \gamma)]$$

$$c + a' = (1 + r)a + \tilde{y}(1, R, z_R, z_U, \gamma)$$

MIGRATE TO URBAN

$$V_{R \rightarrow U} = \max_{a' \geq \underline{a}} u(c, R, x) + \beta \mathbb{E}[V_U(a', z'_R, z'_U, x', e', \gamma)]$$

$$c + a' + m_c = (1 + r)a + \tilde{y}(1, R, z_R, z_U, \gamma)$$

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RECURSIVE FORMULATION: URBAN HOUSEHOLD

urban household stays or migrates to rural:

$$V_U(a, z_R, z_U, x, e, \gamma) = \max \{ V_{U \rightarrow U}, V_{U \rightarrow R} \}$$

STAY IN URBAN

$$V_{U \rightarrow U} = \max_{a' \geq \underline{a}} u(c, U, x) + \beta \mathbb{E}[V_U(a', z'_R, z'_U, x', e', \gamma)]$$

$$c + a' = (1 + r)a + \tilde{y}(e, U, z_R, z_U, \gamma)$$

MIGRATE TO RURAL

$$V_{U \rightarrow R} = \max_{a' \geq \underline{a}} u(c, U, x) + \beta \mathbb{E}[V_R(a', z'_R, z'_U, x', \gamma)]$$

$$c + a' + m_c = (1 + r)a + \tilde{y}(e, U, z_R, z_U, \gamma)$$

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BELIEF DYNAMICS OF URBAN HOUSEHOLDS

1. **Income beliefs:** while in U , households use V with $\gamma = 1$
 - do not anticipate forgetting the true urban wage after returning to R
2. **Migration cost beliefs:** while in U , households use V with $m_c = 0$
 - do not anticipate facing migration cost after returning to R for repeat $R \rightarrow U$ moves
 - allows the stationary equilibrium to hit URPG, urban share moments

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